Pipeline Inspection ESP32-CAM Car

INTRODUCTION:

Pipelines and sewer systems are the backbone of cities. They quietly move water, wastewater, gas and industrial fluids through cities and factories. Often these Pipeline and sewer systems run for years without anyone checking the Pipeline and sewer systems. The problem is, issues like cracks, leaks, blockages and corrosion do not show up away. By the time a problem appears on the surface a lot of damage has already been done to the Pipeline and sewer systems.

Traditional inspection methods are not practical or safe for the Pipeline and sewer systems. They usually require people to enter spaces or use expensive equipment to inspect the Pipeline and sewer systems.

The Pipeline Inspection ESP32-CAM Car is a low-cost robot that can go inside pipes and sewers and show us what is going on inside the pipes and sewers. It uses an ESP32 CAM module, which is a small but powerful computer that can connect to Wi-Fi and has a built-in camera to take pictures of the pipes and sewers. The Pipeline Inspection ESP32-CAM Car drives through the pipes and sewers. It streams video to a mobile app so we can see what is happening inside the pipes and sewers. A special 360-degree camera attachment gives a view of the pipe walls so no cracks or defects in the pipes and sewers are missed.

What makes the Pipeline Inspection ESP32-CAM Car system special is its smart detection features for the pipes and sewers. The Pipeline Inspection ESP32-CAM Car does not just record video of the pipes and sewers. It finds problems like cracks, leaks and blockages in the pipes and sewers. It marks them on the screen so operators can fix the problems with the pipes and sewers right away. Every picture of the pipes and sewers is saved with a date and time stamp creating a log for maintenance teams to fix the pipes and sewers. A toxic gas sensor adds a safety feature alerting operators to dangers like hydrogen sulphide or methane in the pipes and sewers. The Pipeline Inspection ESP32-CAM Car system also estimates the distance to blockages in the pipes and sewers helping engineers plan their work on the pipes and sewers.

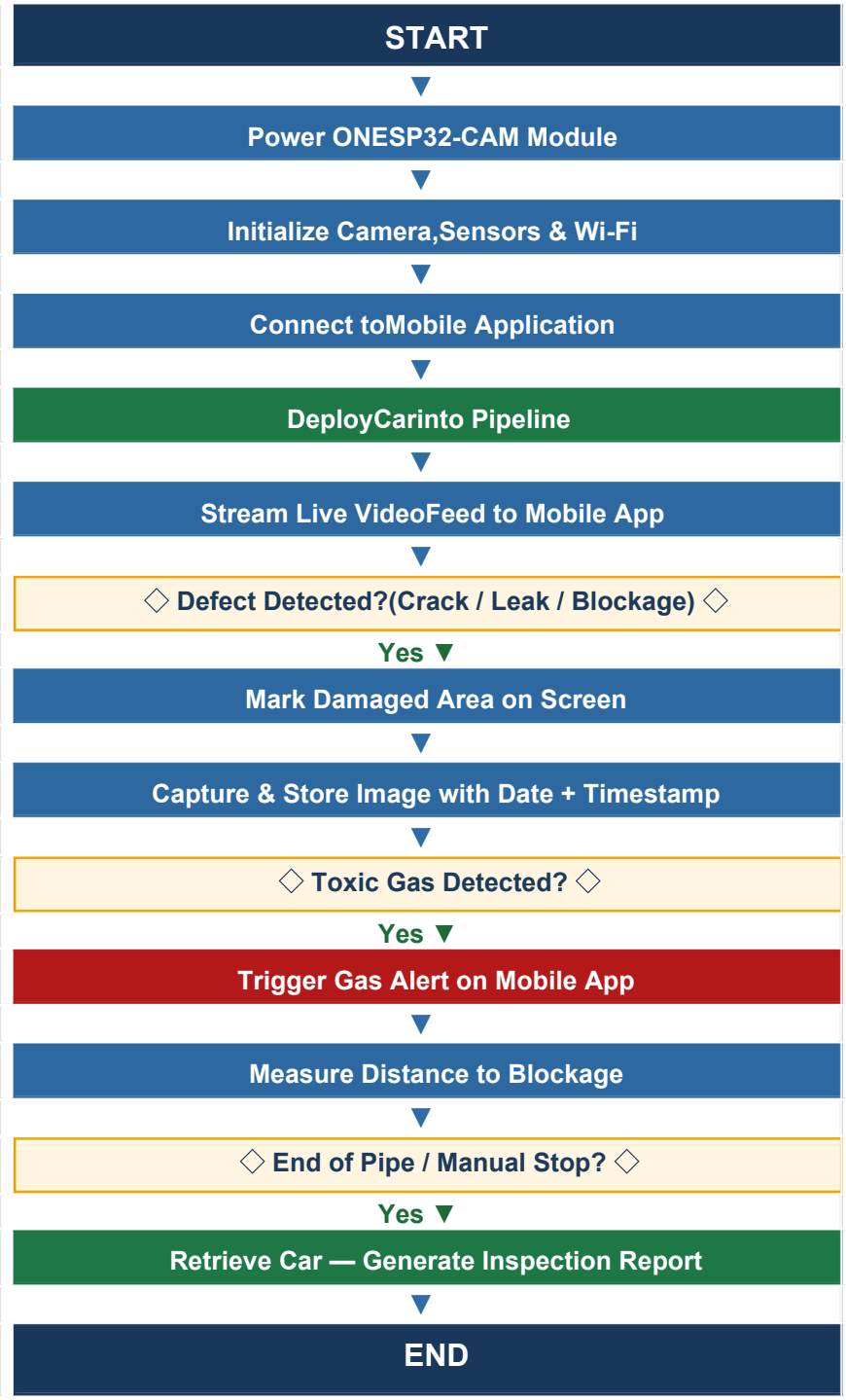
The Pipeline Inspection ESP32-CAM Car is also a way to learn about embedded systems, wireless communication and the Internet of Things (IoT) for the pipes and sewers. It shows how simple components can be combined to solve real-world problems with the pipes and sewers. The Pipeline Inspection ESP32-CAM Car system is designed to be affordable and simple making it accessible to towns, research institutions and students to use on the pipes and sewers. In a world where infrastructure is ageing fast tools like the Pipeline Inspection ESP32-CAM Car could make a difference in how we detect and respond to problems, with the pipes and sewers.

Key System Features

Camera & Vision	Safety & Sensing
• Live video streaming to mobile app • 360° panoramic view of pipe interior • Auto-detect cracks, leakage, blockages • Mark damaged area on screen • Store images with date & timestamp	• Toxicgas sensor (H2S, CH4, CO) • Real-time alert on mobile app • Blockage distance measurement • Low-power, reliable wireless comms • Compact chassis for narrow pipes

SYSTEM FLOWCHART:

Thefollowingflowchartillustrates the complete operational logic of the Pipeline Inspection ESP32-CAM Car, from system initialisation through detection, alerting, and shutdown.



METHODOLOGY:

3.1 HardwareArchitecture

The pipeline inspection car is made up of three parts that work together: the locomotion chassis, the sensing array and the computation and communication core.

The locomotion chassis is basically a frame with wheels that can fit inside sewer and inspection pipes.

It has two or four wheels that are powered by motors.

These motors are controlled by a driver, which lets the pipeline inspection car move forward backward and turn by changing the speed of each wheel.

If the pipe is very rough or wet the pipeline inspection car can use tracks of wheels.

The ESP32-CAM is the brain of the pipeline inspection car system

It has a processor, memory, Wi-Fi and a camera that can take quality pictures and videos.

This module is small. Does not use a lot of power which makes it perfect for the pipeline inspection car that runs on batteries and has to fit in a small space.

The camera has a lens that lets it see a lot of the area inside the pipe.

The pipeline inspection car also has a sensor that can detect gases like the kind that can catch on fire, hydrogen sulphide and carbon monoxide.

These are the dangers found in sewer systems.

The pipeline inspection car has another sensor that uses waves to measure how away things are, which helps the pipeline inspection car find blockages.

The pipeline inspection car also has a light and a buzzer that can tell us what is going on even if we are not connected to the internet.

The pipeline inspection car has all these parts to help it do its job.

The pipeline inspection car is very good, at inspecting pipes.

3.2 Communication and Streaming

The ESP32-CAM operates as a Wifi access point or connects to an existing hotspot, making its video stream available on a local IP address. The mobile application built using MIT App Inventor or a custom Android/iOS interface connects to this address and renders the MJPEG stream in near real time. Control commands for forward, backward, left, right, and stop are sent from the app to the ESP32 via HTTP GET requests or a simple WebSocket channel, keeping latency below 80ms on a stable 2.4 GHz link.

Images flagged by the detection algorithm are saved directly to a microSD card inserted into the ESP32-CAM's onboard slot. Each file is named with a Unix timestamp to ensure chronological ordering and uniqueness. A lightweight HTTP server running on the ESP32 also allows the mobile app to browse and download stored images over the same Wifi connection without requiring a separate internet link or cloud account.

3.3 Defect Detection Algorithm

The way we find defects is done in two steps. The first step happens on the ESP32-CAM. It uses methods like frame differencing and edge detection. These methods do not need a lot of computer power. Can work with the limited memory of the module. When the camera is not moving or is moving slowly it looks at two frames one after the other. If it finds a change in the pixels and strong edges it thinks there might be a problem with the structure.

The second step sends the frames to the device for a closer look. The phone uses a program called TensorFlow Lite that has already been trained to recognize problems like cracks and corrosion in pipes. This program looks at each part of the frame. Says if it is normal cracked, leaking or blocked. We made the program simpler so it can run well on smartphone processors. If the program is very sure it has found a defect it draws a box around the problem on the live video and sends a message to the person operating the camera. The ESP32-CAM and the mobile device work together to find defects, like pipe cracks and corrosion.

3.4 Gas Detection and Alerting

The MQ series sensor outputs an analogue voltage proportional to the concentration of target gases in the surrounding air. This voltage is sampled by the ESP32's ADC at 1 Hz intervals. A software threshold, calibrated against known safe exposure limits defined by occupational health standards, triggers an immediate alert packet to the mobile application when exceeded. The app displays a full-screen warning in red with an audible alarm, allowing the operator to stop and retract the car before any personnel approach the pipe opening. All gas readings are logged alongside the corresponding GPS or pipe-segment reference tag.

3.5 Blockage Distance Measurement

The HC-SR04 ultrasonic sensor emits a 40 kHz pulse and measures the time taken for the echo to return. The ESP32 firmware applies the standard time-of-flight calculation to convert this duration into a distance reading in centimetres. Readings are taken every 500 milliseconds and displayed as a live metric on the mobile app interface. When a blockage is detected — defined as a sustained reading below a configurable minimum clearance threshold — the app logs the estimated distance from the pipe entry point, calculated by integrating wheel encoder pulses or approximating from average speed and elapsed time.

3.6 Power Management

The entire system is powered from a 7.4 V lithium-polymer battery pack, regulated to 5 V for the ESP32-CAM and motors and further stepped down to 3.3 V for the sensors. A power-saving mode in the ESP32 firmware reduces the camera frame rate and Wi-Fi transmission interval when no motion is detected, extending operational endurance to approximately 45–60 minutes per charge cycle depending on pipe conditions and streaming quality settings.

REFERENCES

[1] KPERFORMANCE EVALUATION OF IOT BASED INTELLIGENT CRACK DETECTION SYSTEM FOR CONCRETE BUILDING BLOCKS

[2] AI-Enabled Drone and Sensor Systems for Autonomous Inspection and Maintenance of Underground Utility Networks

[3] Design and fabrication of in-pipe inspection robot,2018.

[4] A modular, multi-sensor crawler robot for adaptive pipeline inspection: design and experimental validation

[5] Design and construction of an inspection robot for the sewage pipes,2017

BIBTEX ENTRIES

The following BibTeX entries may be copied directly into a .bib file for use with LaTeX reference managers such as BibTeX or BibLaTeX.

[1]
```
@article{gond2024performance,
  title={PERFORMANCE EVALUATION OF IOT BASED INTELLIGENT CRACK DETECTION SYSTEM FOR CONCRETE BUILDING BLOCKS},
  author={GOND, ABHISHEK},
  year={2024}
}
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[2]
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@inproceedings{kanimozhi2025ai,
  title={AI-Enabled Drone and Sensor Systems for Autonomous Inspection and Maintenance of Underground Utility Networks},
  author={Kanimozhi, M and Ram, R Saravana and Sabarinathan, S and Natarajan, Eswara Prasath and Jasper, S Kevin and Kaliappan, S},
  booktitle={2025 International Conference on Advances in Next-Gen Computer Science (ICANCS)},
  pages={1--5},
  year={2025},
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}
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[3]
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@article{rakate2018design,
  title={Design and fabrication of in-pipe inspection robot},
  author={Rakate, Akshay B and Navale, Akshaykumar B and others},
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author={Eltwab, Ahmed A Abd and Sameh, Ahmed},
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author={Knedlov{\'a}, Jana and B{\'\i}lek, Ond{\v{r}}ej and S{\'a}mek, David and Chalupa, Petr},
booktitle={MATEC Web of Conferences},
year={2017},
organization={EDP Sciences}
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